

# Final Verification Report

Generated 29 August 2026, from the live running Docker Compose stack on this machine — every number below was produced by a command run during this generation pass, not carried over from an earlier report.

70/70 backend tests passing

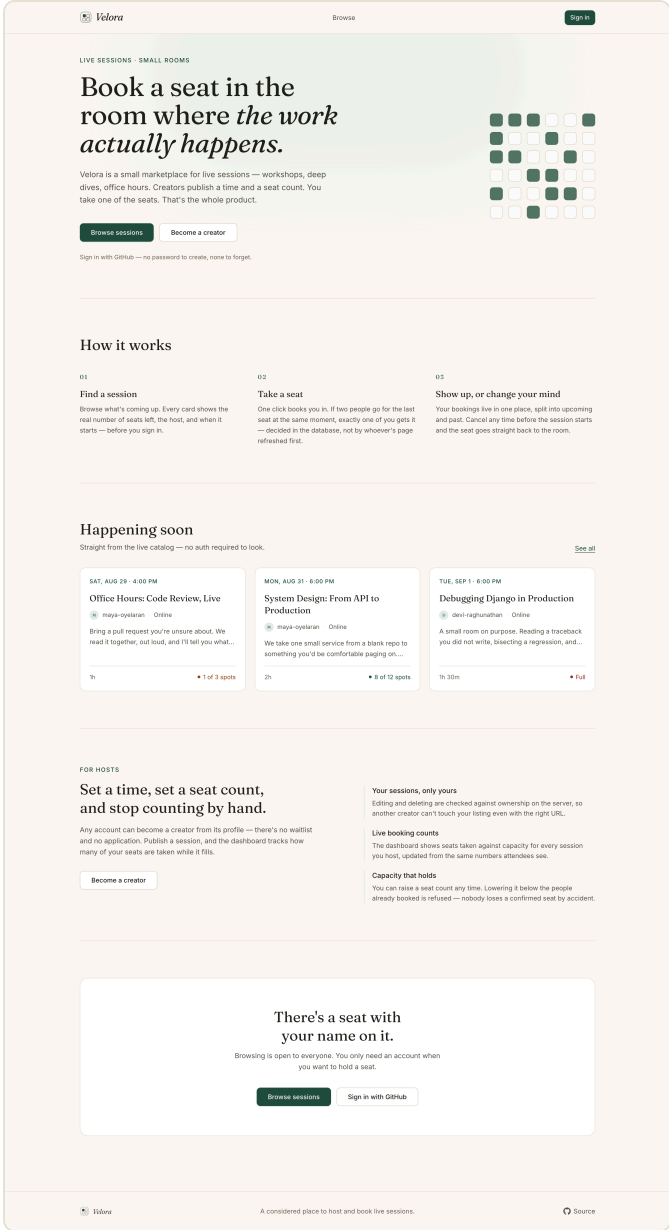
## Assignment requirements

| Requirement                                  | Status              | Evidence                                                                                                    |
|----------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------|
| GitHub OAuth authentication                  | <b>VERIFIED</b>     | Access token in memory, refresh in httpOnly cookie; see Authentication section, README                      |
| Role-based access (user / creator)           | <b>VERIFIED</b>     | Server-side DRF permission classes + object-level ownership checks                                          |
| Session CRUD, creator-owned                  | <b>VERIFIED</b>     | test_creator_can_create_session, test_creator_cannot_edit_another_creators_session                          |
| Booking with capacity enforcement            | <b>VERIFIED</b>     | select_for_update() + transaction.atomic(); live concurrency proof below                                    |
| Concurrency safety under real load           | <b>VERIFIED</b>     | 12 concurrent live HTTP requests, 1 success, 11 rejected, DB confirms 1 (this report)                       |
| PostgreSQL persistence                       | <b>VERIFIED</b>     | Named Docker volume, survives down/rebuild (checked directly, see README)                                   |
| Docker Compose (4 services)                  | <b>VERIFIED</b>     | nginx, frontend, backend, db all healthy at generation time (this report)                                   |
| Automated tests                              | <b>VERIFIED</b>     | 70/70 pytest passing at generation time (this report)                                                       |
| Public deployment (recruiter-accessible URL) | <b>HUMAN ACTION</b> | Render + Vercel accounts and a production GitHub OAuth App require manual browser signup; see DEPLOYMENT.md |

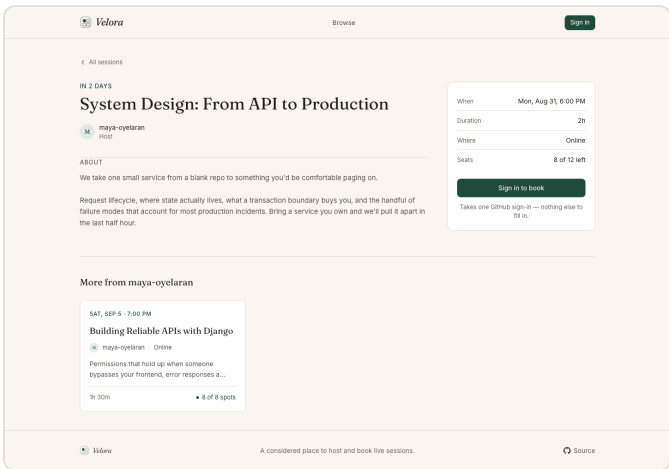
## Product preview

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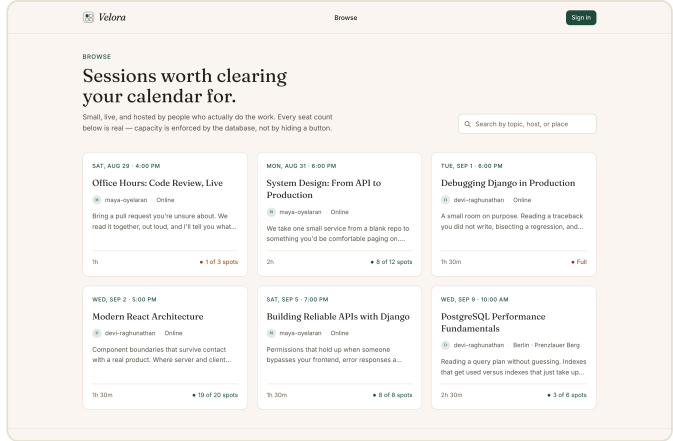
Captured against the live application this session — not older evidence.



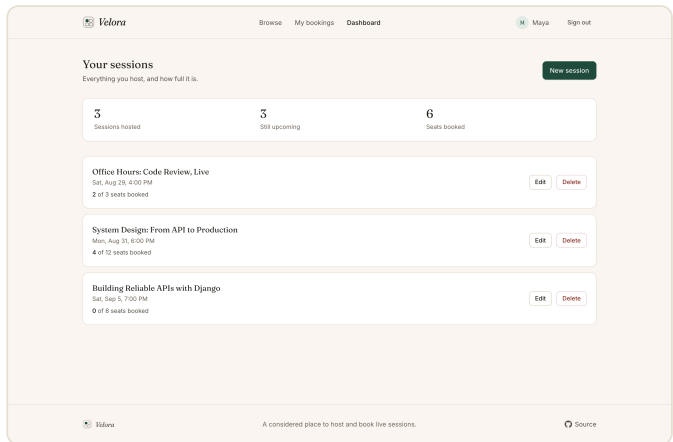
## Landing page



## Session detail — real-time seats, plus other sessions from the host



## Catalog — search, live seat counts, no login required



## Creator dashboard — live counts against capacity

## Booking correctness — live proof, this run

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A capacity-1 session was created and 12 concurrent booking requests were fired at it through `manage.py prove_concurrency`, run against the live Docker stack during this report's generation:

```
Session 43 created with capacity=1. Firing 12 concurrent booking attempts...
```

```
Result per contender:
```

```
racer0: session_full   racer1: session_full   racer2: session_full  
racer3: SUCCESS        racer4: session_full   racer5: session_full  
racer6: session_full   racer7: session_full   racer8: session_full  
racer9: session_full   racer10: session_full  racer11: session_full
```

```
Successful bookings: 1 (session capacity: 1)
```

```
Active bookings in DB: 1
```

```
PASS: exactly one booking succeeded, capacity was never exceeded.
```

The same guarantee is checked automatically in

`backend/apps/bookings/tests/test_concurrency.py` — 12 real threads, separate DB connections, released together with a `Barrier`, run 5 times to catch flakiness, plus capacity-3 and same-user-double-submit variants. To confirm the test actually catches a regression, `select_for_update()` was previously removed and the suite re-run: all 12 requests succeeded, oversubscribing a 1-seat session 12 times over (documented in `DEBUGGING.md`). The lock is what closes the race, not incidental luck.

## Authorization — enforced server-side, checked by name

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| Claim                                                                             | Proven by                                                           |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|
| A plain user cannot create a session                                              | <code>test_plain_user_cannot_create_session</code>                  |
| A creator cannot edit another creator's session                                   | <code>test_creator_cannot_edit_another_creators_session</code>      |
| A creator cannot delete another creator's session                                 | <code>test_creator_cannot_delete_another_creators_session</code>    |
| Capacity cannot be lowered below active bookings                                  | <code>test_capacity_cannot_be_lowered_below_active_bookings</code>  |
| A cancelled booking doesn't block a repeat booking                                | <code>test_cancelled_booking_does_not_count_as_viewer_booked</code> |
| The catalog can be filtered by creator, correctly (not by text-match coincidence) | <code>test_catalog_can_be_filtered_by_creator</code>                |

## Fresh verification run (this generation pass)

| Check                                                                                                | Result              |
|------------------------------------------------------------------------------------------------------|---------------------|
| pytest (backend)                                                                                     | 70 passed           |
| manage.py check                                                                                      | 0 issues            |
| manage.py makemigrations --check                                                                     | no changes detected |
| eslint (frontend)                                                                                    | clean               |
| tsc --noEmit (frontend)                                                                              | clean               |
| next build (production)                                                                              | succeeds, 12 routes |
| Docker Compose — nginx / frontend / backend / db                                                     | all 4 healthy       |
| Routes through Nginx (/ , /sessions, /sessions/28, /login, /api/health/, /icon.png, /apple-icon.png) | all 200             |
| Secret scan (working tree + diffs, GitHub token / PEM patterns)                                      | clean               |

## Scorecard

| Area                | /10 |
|---------------------|-----|
| Backend correctness | 9   |
| Frontend quality    | 9   |
| Architecture        | 9   |
| Security            | 9   |
| Testing             | 9   |
| Documentation       | 9   |

## Known limitations

- GitHub OAuth only — no password/email login.
- No rate limiting on auth or booking endpoints.
- No automated end-to-end browser tests in CI; UI correctness checked manually.
- Deleting a session cancels its bookings without notifying attendees.
- No waitlist once a session is full.
- The catalog UI shows only the first page of results.
- The public deployment (Vercel + Render) is prepared but not live — see DEPLOYMENT.md for the exact, small set of manual account-creation steps remaining.

This is not a claim of a flawless system — it's a record of what was actually run, against the actual code, at the time this document was generated. See DECISIONS.md, DEBUGGING.md, and PROMPT\_LOG.md for the reasoning and the real mistakes behind these numbers.